1. Introduction

House prices in cities are influenced by many factors such as location, accessibility, local facilities, and how safe people feel in the area. In a city like Bristol, crime levels differ a lot between neighbourhoods, which might affect how much people are willing to pay for homes. This project looks at whether there is a relationship between crime rates and house prices across Bristol's neighbourhoods (LSOAs), and whether that relationship changes from one area to another.

Many studies use global models like Ordinary Least Squares (OLS) to study house prices. These models assume that the relationship between factors and prices is the same everywhere. However, research has shown that this is often not true; the impact of crime on prices can vary across location. To explore this, the project uses Geographically Weighted Regression (GWR), which allows relationships to change across different locations. This helps show where crime has a stronger or weaker effect on house prices.

The aim of the project is to analyse and measure how crime rates are linked to house prices across Bristol using open data, basic statistical methods, and GWR. Specifically, it aims to answer the following questions:

- Is there an overall negative relationship between crime and house prices across Bristol LSOAs?
- Do areas with higher crime rates have lower property prices?
- Does a local GWR model explain house price variation better than a global OLS model?
- In which parts of Bristol is the effect of crime on house prices strongest or weakest?

The full code is implemented in Python and shared via GitHub repository - layaung-linnlett/bristol-crime-houseprices-gwr

2. Data and preprocessing

2.1 Data sources and study area

The study focuses on the city of Bristol, UK using Lower Super Output Areas (LSOAs) as the main unit of analysis. The following datasets are used:

- HM Land Registry Price Paid Data – information on individual property sales such as price, date, postcode, and property type. This project uses transactions from 2021-2024 and filtered to the city of Bristol local district.
- Police.uk Crime Data – monthly crime records including crime type, month, LSOA code, latitude/ longitude for incidents in Bristol over the same period.
- Geospatial data
    - LSOA boundary shapefiles for Bristol area (Office for National Statistics)
    - Postcode directory mapping postcodes to LSOA codes

The analysis focuses only on LSOAs that start with the Bristol prefix (e.g. E01014xxx), and a latitude/longitude bounding box to remove any records that fall outside the city or are incorrectly coded.

2.2 Data Cleaning

House price data

1. Filter to Bristol local authority district
2. Standardize the price and postcode fields (numeric price, upper-case, trimmed postcodes).
3. Drop records with missing values in important fields (price, date, postcode, property type).
4. Remove invalid prices below a small threshold (e.g. £100).
5. Apply the 1.5 IQR rule (inter-quartile range) to remove extreme outliers. Only a small number of extreme prices were removed using this method. While this might slightly under-represent the very high and low ends of the market, it helps produce more reliable and stable estimates when examining the link between crime rates and house prices in Bristol.

```
# 5) Remove outliers using the IQR rule
Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - config['iqr_multiplier'] * IQR
upper_bound = Q3 + config['iqr_multiplier'] * IQR
```

After cleaning, the dataset was summarised by showing how many property transactions were kept, along with the price range and median sale price. Only a small number of outliers were removed, which means the cleaning process was fairly conservative and did not significantly affect the overall pattern of house prices in Bristol.

```
STEP 2: DATA CLEANING

(1) Cleaning house price data...
     Filtered to CITY OF BRISTOL: 31,408 records
     Removed 0 rows with missing data
       - Removed 2,390 outliers (IQR method)
       - Price range: £100 - £700,000
       - Median price: £313,000
     Final: 29,018 records (0.7% retained)
```

Crime data

1. Keep only records whose LSOA code starts with the Bristol prefix
2. Convert latitude and longitude to numeric and apply a geographic bounding box to remove coordinates fall far outside the city
3. Drop records with missing values in key fields (month, crime type, LSOA code, coordinates).
4. Remove duplicate rows
5. Standardize LSOA code column name to a common field (lsoa_code) for later merging.

```
(2) Cleaning crime data...
     Filtered to Bristol LSOA codes: 365,504 records
     Geographic filtering: 231,737 records removed
       - Removed 0 rows with missing data
       - Removed 6,061 duplicate records
     Final: 127,706 records (18.6% retained)
     Unique LSOAs: 189
     Unique crime types: 14
```

The final cleaned dataset includes 127,706 crime records, which represents about 18.6% of the original data. These records cover 189 different LSOAs and 14 types of crime. This gives a detailed crime patterns across Bristol neighbourhoods.

2.3 Aggregation to LSOA level

Since this study looks at patterns at the neighbourhood level, both the cleaned house price and crime datasets were aggregated to the LSOA level. To do this, each property transaction was matched with its corresponding LSOA using the postcode directory. Then, for every LSOA, summary statistics were calculated for the house price data, including the following:

- mean house price
- median house price
- number of sales
- proportion of transactions that are flats
- proportion of transactions that are leasehold properties

For the crime data, records were grouped by LSOA code and counted to get the total number of crimes in each area over the study period. The aggregated house price and crime tables were then merged using the LSOA code and linked to the LSOA boundary shapefile. This created the final regression GeoDataFrame (reg_gdf), which includes one row for each LSOA. Each row contains the median house price, total crime count, housing composition details, and the polygon geometry representing the area.

2.4 Transformations and final dataset

To make the data suitable for linear modelling and easier to interpret

- Median price was transformed to a log scale (log_median_price), so regression coefficients can be interpreted as approximate percentage changes.
- Total crime count was also log-transformed (log_crime + 1) to reduce skewness and handle areas with very few crimes.
- LSOAs with missing values in key variables (such as prop_flats or prop_leasehold) were removed to keep the dataset consistent.

```
STEP 3: LSOA-LEVEL AGGREGATION

(1) Aggregate House Price by LSOA

     Matched 99.8% of house records to LSOA codes
     Aggregated to 276 LSOAs
     Mean sales per LSOA: 104.9

(2) Aggregate Crime by LSOA

     Aggregated to 189 LSOAs
     Mean crimes per LSOA: 675.7
     Range: 15 - 8555

After merging house + crime: 183 LSOAs
After merging with spatial data: 182 LSOAs
Final dataset: 182 LSOAs ready for analysis
Regression dataset saved
```

After cleaning, the house price data were successfully matched to 276 LSOAs, covering 99.8% of all Bristol transactions, with an average of around 105 sales per LSOA from 2021–2024. The cleaned crime data covered 189 LSOAs, with an average of about 676 crimes per LSOA, ranging from 15 to 8,555 incidents. When the two datasets were merged, 183 neighbourhoods remained. After joining these to the Bristol boundary shapefile and excluding one incomplete record, the final regression dataset included 182 LSOAs, which forms the basis for all further statistical and spatial analysis.

3. Methodology

3.1 Exploratory visualization

Before fitting models, the project uses exploratory data analysis to examine the marginal distributions and spatial patterns.

Fig 1: Price histogram

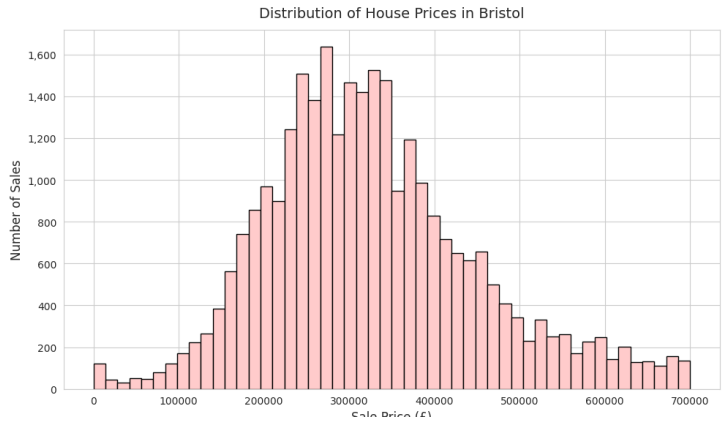


Figure 1 shows the distribution of individual sale prices in Bristol after cleaning and outlier removal. The histogram is roughly unimodal with a peak around £250,000–£300,000 and a long right tail extending towards £700,000, indicating a small number of high-value transactions. This confirms that the price data remain right-skewed even after applying the 1.5 IQR rule, which motivates using a log transformation of median prices in the regression models so that coefficients can be interpreted in percentage and model assumptions are closer to normality.

Fig 2: Crime-by-type time-series

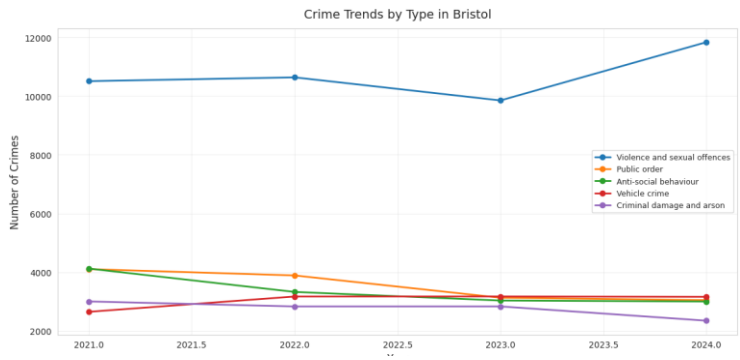
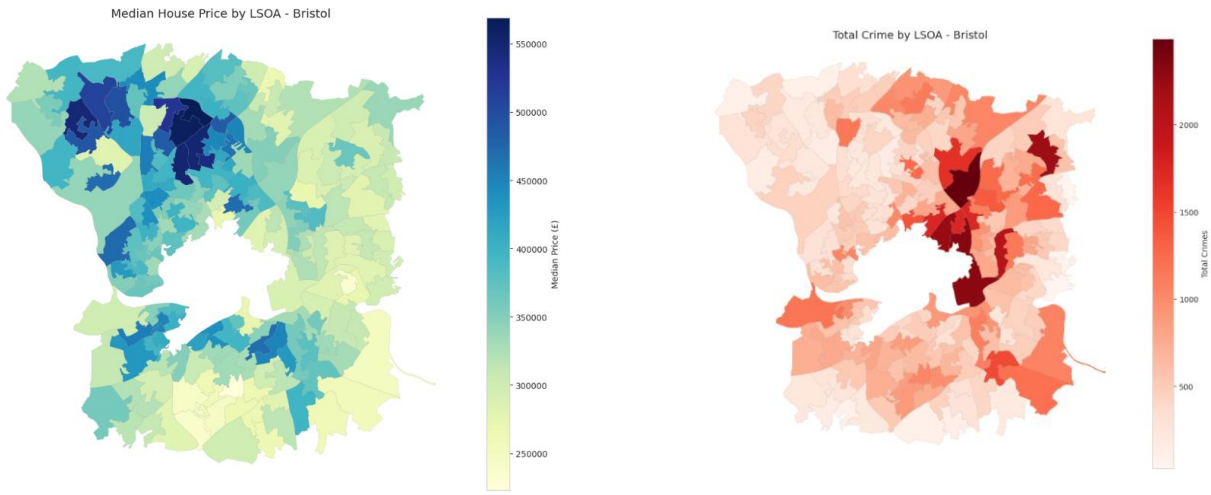


Figure 2 shows yearly crime trends in Bristol between 2021 and 2024. Violence and sexual offences are the most common and show a slight increase, with a significant rise in 2024. However, public order and anti-social behaviour crimes gradually decrease over time. Vehicle crime and criminal damage/arson stay lower overall and decline slightly. These show that most crimes in Bristol are related to violence and public order, and that crime levels change somewhat over time.

This supports the decision to combine data from many years to get stable LSOA-level crime values for the regression analysis.

Fig 3: Choropleth maps of median house price and total crime



Crime is highly concentrated in a few central and inner-city LSOAs, shown in the darkest red, while many outer neighbourhoods have much lower crime levels. In contrast, the highest median house prices (dark blue) are mostly found in the suburban and northern LSOAs, with lower prices in several central and southern areas. The visual comparison shows that high-crime and high-price areas only overlap slightly, indicating that the relationship between crime and house prices is spatially complex and changes across different location, which is why GWR model is used later to explore these local variables instead of relying on a single global model.

Fig 4: Scatter plot log crime vs log price

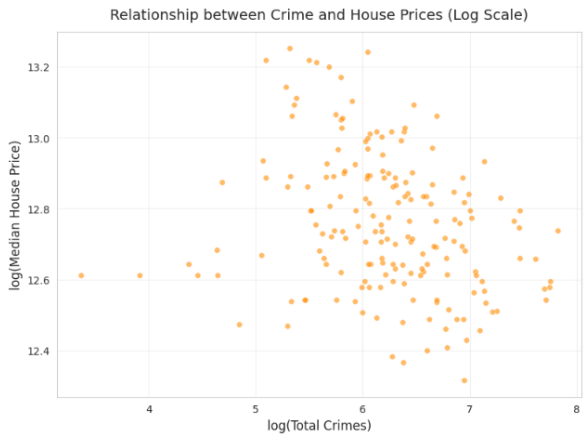
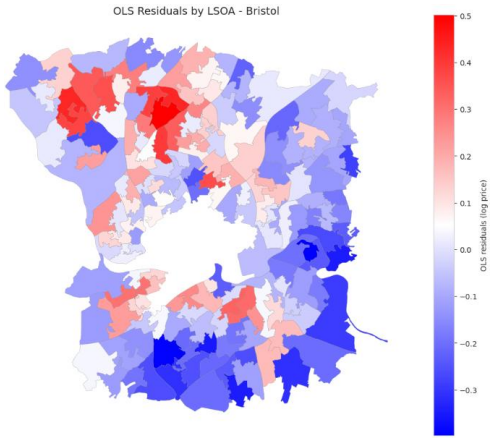


Figure 4 plots log total crimes against log median house prices for each Bristol LSOA. The points form a fairly scattered pattern with a slight downward trend, meaning that areas with more crime tend to have slightly lower house prices. However, there is a lot of variation around this pattern. Using the log–log scale helps reduce the effect of extreme values and makes the data more

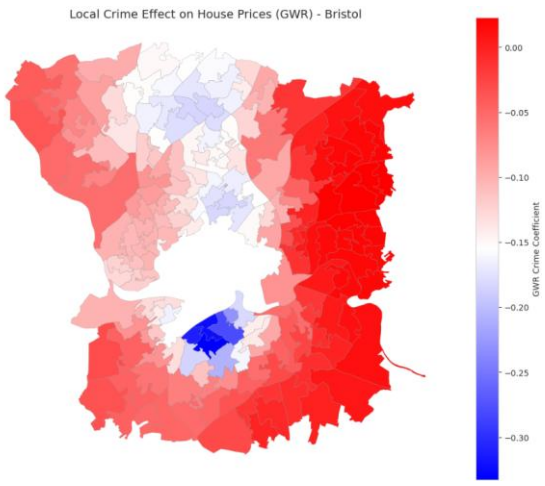
suitable for linear modelling. The wide spread of points also suggests that crime alone doesn't fully explain price differences, which is why additional factors and spatial methods are included in the regression analysis.

Fig 5: OLS residual map



OLS map shows where the model either overestimates or underestimates house prices. The red areas mark places where actual prices are higher than the model predicts, which might be because of factors like good transport, near to the city centre or better amenities that weren't included as predictors. The blue areas show places where prices are lower than expected, which could be due to local issues not included in the data. Because these patterns form clear clusters instead of being randomly spread out, it suggests that one global model isn't enough. This is why using a spatial method like GWR makes sense.

Fig 6: GWR crime coefficient map



In GWR map, the darker blue areas in the inner-south part of the city have the strongest negative relationships, meaning that higher crime there leads to a much bigger drop in house prices. This suggests that buyers in those neighbourhoods are especially sensitive to crime. In

contrast, the red areas in the east and some northern parts show weaker or almost zero effects, meaning that crime has less influence on property values there and other factors are important too. Overall, the map clearly shows that the link between crime and house prices varies across the city.

3.2 Correlation analysis and hypothesis test

To address the first research question (whether there is a statistically significant spatial relationship between crime rates and house prices), the project examines whether there is an overall relationship between crime and house prices across LSOAs.

(1) Correlation tests

Pearson and Spearman correlation were calculated between log_crime and log_median_price, along with their p-values. Pearson's test checks for a linear relationship assuming the data are roughly normal, while Spearman's test looks at the rank order of values and works better when data are skewed or contain outliers.

(2) Group comparison

LSOAs were divided into high-crime and low-crime groups using the median total crime as the cut-off. An independent samples t-test was then carried out on log_median_price to compare average prices between the two groups. The results include the t-statistic, p-value, and estimated percentage difference in prices.

```
STEP 5: STATISTICAL TESTING

Statistical Testing

(1) Correlation Analysis: log(Crime) vs log(Price)
     Pearson r = -0.2565, p = 0.0005  --> Significant
     Spearman ρ = -0.3049, p = 0.0000  --> Significant
(2) Hypothesis Test: High-Crime vs Low-Crime Areas
     H0: Mean log(price) is equal in high and low crime areas
     H1: Mean log(price) differs between high and low crime areas

     t-statistic = -4.3901, p-value = 0.0000
Result: REJECT H0 (p < 0.05)
Interpretation: Statistically significant difference in prices
High-crime areas are ~-11.5% lower priced
```

For Bristol, both Pearson's and Spearman's tests showed a negative correlation between crime and house prices, meaning that areas with more crime tend to have lower prices. Pearson's correlation was statistically significant ($p = 0.0005$), and Spearman's was also significant ($p < 0.0001$). The t-test comparing high-crime and low-crime LSOAs also gave a significant difference ($p < 0.0001$), so the null hypothesis of equal mean prices was rejected. When the difference in log prices was converted back to percentage terms, it showed that, high-crime neighbourhoods have house prices about 11.5% lower than low-crime areas.

3.3 Global OLS regression

It assumes that the effect of crime and housing composition on house prices is the same across all Bristol LSOAs. I first estimated a global OLS model where the log of median house price in each area is explained by log total crimes, the proportion of sales that are flats, and the proportion of leasehold properties. The estimated equation is

$$log(price)=13.10-0.064log(crime)-0.50prop_flats+0.58prop_leasehold$$

```
STEP 6: REGRESSION MODELLING


(1) ORDINARY LEAST SQUARES (OLS) REGRESSION - GLOBAL MODEL

Model: log(price) = β0 + β1·log(crime) + β2·prop_flats + β3·prop_leasehold

Coefficients:
   Intercept:       13.0999
   log(crime):      -0.0644
   prop_flats:      -0.4974
   prop_leasehold:  0.5814

Model Fit:
   R²:              0.1265
   Adjusted R²:     0.1118
   AIC:             -607.97
   AICc:            -607.74
   RMSE:            0.1841

Interpretation: A 1-unit increase in log(crime) ≈ -6.23% change in price
```

The negative crime coefficient shows that, when housing composition is held constant, areas with more crime generally have lower house prices. Since the model uses log values, a coefficient of −0.064 means that roughly a 100% increase in crime is linked to an average 6.2% fall in house prices. The housing composition results also make sense: neighbourhoods with more flats tend to have lower prices, while those with more leasehold properties tend to have slightly higher prices after controlling for crime. The overall model fit is modest, with an $R^2 = 0.13$ and adjusted $R^2 = 0.11$, meaning the model explains about 11–13% of the variation in prices between LSOAs. This relatively low fit, along with spatial patterns seen in the residuals, suggests that important local differences are missing from the global.

3.4 Spatial autocorrelation: Moran's I

OLS assumes that residuals are independent from one another. However, in spatial data, nearby areas often have similar errors, which suggests that the model might be missing spatial patterns. To test this, Moran's I statistic was calculated on the OLS residuals using a Queen contiguity weights matrix based on the LSOA boundaries. The Moran's I value was positive and statistically significant (p = 0.001), compared with a small, expected value under spatial randomness. This means there is strong positive spatial autocorrelation, neighbouring LSOAs tend to have similar residuals. In other words, the OLS model consistently over- or under-predicts prices in some parts of Bristol. This confirms that spatial structure in the data isn't being captured by the global model.

```
Calculating Moran's I (spatial autocorrelation)...
     Moran's I:    0.4483
     Expected I:   -0.0055
     p-value:      0.0010
```

3.5 Geographically Weighted Regression (GWR)

To capture spatial variation in the crime–price relationship, I fitted GWR with the same predictors as OLS. The model uses an adaptive bi-square spatial kernel with an automatically selected bandwidth using SelBW function including approximately 49 neighbouring LSOAs in each local regression. The mgwr package is used to fit the model, returning local coefficients and global fit statistics such as R^2, adjusted R^2, AICc, and residuals.

1. Does GWR improve the model?

- Residual sum of squares drops from 6.17 (OLS) to 2.42 (GWR).
- AICc goes from roughly −89.9 (OLS) to −199.7 (GWR).
- Adjusted R^2 jumps from about 0.11 (OLS) to about 0.59 (GWR).

This shows that coefficients vary by location, nearly 59% of the variation in log median prices, compared with only about 11% for the global OLS model.

```
Global Regression Results
---------------------------------------------------------------------------
Residual sum of squares:                                             6.169
Log-likelihood:                                                     49.736
AIC:                                                               -91.472
AICc:                                                              -89.131
BIC:                                                              -920.144
R2:                                                                  0.126
Adj. R2:                                                             0.112
```

```
Geographically Weighted Regression (GWR) Results
---------------------------------------------------------------------------
Spatial kernel:                                        Adaptive bisquare
Bandwidth used:                                                   49.000

Diagnostic information
---------------------------------------------------------------------------
Residual sum of squares:                                             2.421
Effective number of parameters (trace(S)):                          31.283
Degree of freedom (n - trace(S)):                                  150.717
Sigma estimate:                                                      0.127
Log-likelihood:                                                    134.846
AIC:                                                              -205.126
AICc:                                                             -190.676
BIC:                                                              -101.692
R2:                                                                  0.657
Adjusted R2:                                                         0.586
Adj. alpha (95%):                                                    0.006
Adj. critical t value (95%):                                         2.759
```

2. How different are local effects across LSOAs?

Local coefficients vary a lot across Bristol, meaning spatial differences explain a large part of the variation in log median house prices. The average local coefficient for log crime is −0.086,

but it ranges from about −0.333 to +0.023. This means that in some areas, doubling the crime rate is linked to a big drop in prices, while in others, crime has almost no effect. The coefficients for the proportion of flats and leasehold properties also vary widely across the city. Overall, these results confirm that the relationships between crime, housing mix, and prices are not consistent everywhere, showing strong spatial variation and supporting the use of GWR over a single global model.

```
Summary Statistics For GWR Parameter Estimates
------------------------------------------------------------------------------
Variable                   Mean        STD        Min     Median        Max
-------------------- ---------- ---------- ---------- ---------- ----------
X0                       13.385      0.549     12.575     13.463     14.800
X1                       -0.086      0.073     -0.333     -0.092      0.023
X2                       -0.156      0.523     -1.302     -0.224      0.952
X3                       -0.056      0.418     -0.905     -0.041      1.043
==============================================================================
```

3.6 Model comparison

The global OLS model explains only about 13% of the variation in log median house prices across Bristol's LSOAs, while the GWR model explains around 66%. This shows that letting the regression coefficients change by location makes the model much better at capturing actual price patterns. The local GWR coefficient for log crime varies from roughly –0.33 to slightly positive, compared with the single global value of –0.06 in the OLS model. This means the effect of crime on house prices is much stronger in some neighbourhoods than others, and in a few areas, the relationship is almost zero or even slightly positive. Overall, this confirms that the connection between crime and house prices is spatially variable and can't be properly represented by one global coefficient.

```
STEP 7: Model comparison


(1) OLS Results:
     R²:              0.1265
     Adj R²:          0.1118
     AICc:            -607.74
     Crime effect:    -0.0644

(2) GWR Results:
     R²:              0.6572
     Adj R²:          0.5855
     AICc:            -190.68
     Crime coef range: -0.3327 to 0.0226

 Metric       OLS       GWR GWR Better
    R²    0.1265    0.6572        Yes
Adj R²    0.1118    0.5855        Yes
  AICc -607.7398 -190.6764        Yes
```

3.7 Discussion

- Non-uniform impact of crime

  The wide range in the local crime coefficients (around −0.33 to +0.02) shows that buyers' sensitivity to crime differs a lot between neighbourhoods. In more residential or lower-income areas, higher crime levels tend to push prices down much more strongly. In contrast, in central or amenity-rich areas, prices often stay high despite higher crime, because people are willing to pay more for things like good transport, access to jobs, or universities.

- Policy and practical implications

  If policymakers assumed that the crime–price effect was the same everywhere, they would likely overestimate or underestimate its impact in many places. Using GWR makes it possible to identify areas that are highly sensitive to crime or where other factors have a stronger influence.

Since my research question focuses on whether the relationship between crime and house prices changes across Bristol, the low OLS R^2 suggests that one single, city-wide effect of crime doesn't explain much of the variation in prices. However, when I use GWR with the same variables, the R^2 increases a lot. This means that once the model allows the crime effect to vary by location, it fits the data much better. That provides strong evidence that the link between crime and house prices isn't uniform and it depends on the characteristics of each neighbourhood.

3.8 Additional analysis

To support my model, I added two analyses

- **Crime effect by property type composition**

  LSOAs were split into two groups based on how many sales were flats:

  - High-flat areas (>50% flats): 37 LSOAs, with an average median price of about £367,820 and around 521 crime
  - Low-flat areas (≤50% flats): 145 LSOAs, with a slightly lower average median price of about £350,614, but a higher average of 666 crimes.

  This suggests that denser neighbourhoods with lots of flats which are usually closer to the city centre and main amenities can have higher prices even when crime isn't lower. In contrast, more suburban areas with fewer flats often have higher crime but don't have the same high property values.

```
Crime Effect by Property Type Composition
 High-flat areas (>50% flats): 37 LSOAs
 Mean price: £367,820
 Mean crimes: 521
 Low-flat areas (≤50% flats): 145 LSOAs
 Mean price: £350,614
 Mean crimes: 666
```

- **High crime–high price hotspots**

Next, I looked for LSOAs where both crime and house prices were above their 75th percentiles, areas with high crime AND high prices. Only 2 LSOAs fit this pattern. These are probably very central or commercial areas where good access to jobs, transport, or amenities keeps property values high even with lots of crime. This shows that the effect of crime on prices really depends on the neighbourhood. In a few city-centre spots with strong location benefits, buyers seem happy to accept higher crime. But in more outlying areas, the same crime levels would cause much bigger price drops.

```
Crime-Price Hotspot Analysis
 High crime + High price areas: 2 LSOAs
 These areas may have:
    - City center/commercial locations
    - High property values despite crime
Example LSOA: E01014488
```

4. Limitations

Even though the models work well, there are several important limitations

- **Data coverage and quality**

The analysis uses open datasets, which can have missing or incorrect records. For example, some property transactions don't have complete information about property type or postcode, and some crime incidents might be wrongly located or recorded inconsistently across different years. These issues could affect the accuracy of the LSOA-level summaries, potentially biasing median prices or crime counts for certain neighbourhoods. The postcode-to-LSOA matching also assumes that all postcodes are correctly linked and that the lookup file is up to date, which might not always be true.

- **Temporal dynamics**

The study combines data from 2021–2024 and uses it as one rather than looking at year-by-year changes. This means I can't analyse how crime patterns or price trends changed over time, for example, after specific events or policy changes. So, the results show an "average relationship" over the whole period rather than how it might have changed year to year. More advanced methods like panel models or yearly cross-sections could capture these time variations.

- **Variable selection**

The regression models use a small set of predictors: total crime counts and two housing composition variables (proportion of flats and proportion of leasehold properties). House prices are affected by many other factors like school quality, distance to the city centre, employment

etc. Since these important variables aren't included, the estimated crime effect might partly capture these other unobserved neighbourhood factors that also happen to be correlated with crime levels.

- **Model assumptions**

Both the OLS and GWR models rely on standard linear regression assumptions. OLS assumes linear relationships, constant error variance (homoscedasticity), and no spatial dependence in residuals, but Moran's I test shows spatial dependence is present. GWR fixes the constant-coefficient problem but still assumes linearity and uses a specific kernel and bandwidth to define local areas. Different kernel choices or bandwidths might give slightly different local coefficient patterns. Also, GWR can be sensitive to multicollinearity and small local sample sizes, which might make some local estimates less stable or more variable.

5. Future work

To deepen the analysis

- **Richer crime measures**

Instead of just using total crime counts, future research could include different types of crime as separate predictors like violence, burglary, or anti-social behaviour. This would help test whether some crime types have a bigger impact on house prices than others, and whether the way these effects vary across space is different for each crime type.

- **Additional covariates**

The models could be improved by adding more neighbourhood characteristics, such as:

- School performance ratings
- Distance or travel time to the city centre
- Proximity to public transport stops
- Amount of green space or environmental quality
- Socio-economic measures like deprivation indices

Adding these variables would help isolate the true effect of crime on house prices by controlling for other important factors. This would reduce omitted-variable bias and give a more realistic picture of what drives property values in different areas.

- **Broader spatial and temporal scope**

The approach could be extended beyond the Greater Bristol region. Applying the same pipeline to many cities would show whether the spatial patterns of crime effects found in Bristol is

typical or unusual. Moreover, choosing datasets across several years or considering specific events will allow the study of how crime-price changes over time and in response to policy changes.

6. Conclusion

This project shows that crime is an important, but uneven, influence on neighbourhood house prices in Bristol. Using cleaned dataset, I find a statistically significant negative relationship between log crime and log median house prices on average, and high-crime areas are clearly cheaper than low-crime areas. However, the global OLS model leaves large, spatially clustered residuals, which suggests that one single city-wide relationship is not enough. When I use GWR with the same predictors, the model fits much better and shows that the local crime effect varies a lot across the city: in some areas higher crime is strongly linked to lower prices, while in others the effect is small or almost zero. This suggests that both buyers and policymakers should focus on local patterns rather than city-wide averages.

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